

# Tianyu Zhang

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[zhangty23@mails.tsinghua.edu.cn](mailto:zhangty23@mails.tsinghua.edu.cn)

| (+86) 186-1542-3591

| [Github](#)

## RESEARCH INTERESTS

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My research focuses on **Internet measurement and cyberspace situational awareness**, with an emphasis on **Web PKI observability**. I build Internet-wide measurement and analysis tools for SSL/TLS certificates and study security issues in certificate issuance, such as certificate misissuance in ACME protocol (*WWW 2025*).

## EDUCATION

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### Tsinghua University

*Doctor of Engineering (D.Eng.) in Electronic Information*

Sep. 2023 - Jun. 2028 (expected), Beijing, China

*Institute for Network Sciences and Cyberspace; Internet Architecture Laboratory*

- Advisors: Prof. Han Zhang and Prof. Jilong Wang
- Research focus: Web PKI measurement, cyberspace situational awareness

### University of Illinois Urbana-Champaign

*Bachelor's Degree in Computer Engineering (Dual-Degree Program)*

Sep. 2019 - May 2023, Urbana, IL, USA

### Zhejiang University

*Bachelor's Degree in Computer Engineering (Dual-Degree Program)*

Sep. 2019 - Jun. 2023, Haining, Zhejiang, China

## PUBLICATIONS

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- **ACME++: A Secure Authorization Mechanism for ACME Clients in the Web PKI Ecosystem**  
**Tianyu Zhang**, Han Zhang, Yunze Wei, Yahui Li, Xingang Shi, Jilong Wang, and Xia Yin  
*The Web Conference (WWW)*, 2025
- **GlassMiner: Mining Looking Glass Services via Structure-Semantics Fusion for Web Observability**  
Yunze Wei, Xingang Shi, Han Zhang, **Tianyu Zhang**, Yahui Li, and Xia Yin  
*The Web Conference (WWW)*, 2026
- **Providing Efficient and Robust ISP-Centric Intrusion Detection Service with Programmable Switches**  
Han Zhang, Xuefeng Liu, Linqiang Qian, Guyue Liu, **Tianyu Zhang**, Kaiyang Zhao, Yantu Tong, Zeji Xiao, Dongbiao He, Yahui Li, Ke Ruan, Jilong Wang, Yongqing Zhu, and Xia Yin  
*ACM SIGCOMM*, 2026

## RESEARCH EXPERIENCE

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## **Tsinghua University, Internet Architecture Laboratory and CNPT Lab**

*Graduate Research Assistant, DeepShield Project*

Sep. 2023 - Present, Beijing, China

- Designed and developed an Internet-wide SSL/TLS certificate database which integrated approximately **70 million actively scanned certificates** and **1.1 billion Certificate Transparency log certificates** into a unified measurement and analysis platform.
- Built certificate-asset relationship graphs and security-analysis capabilities for certificate content analysis, forged-certificate detection, and tens of millions of SSL/TLS certificate asset profiles.
- Identified a cache-related vulnerability in the ACME certificate issuance protocol and developed the **ACME cache-reuse attack**.
- Validated the attack's impact in the Let's Encrypt production environment; this work led to the ACME++ publication at WWW 2025.

## **University of Illinois Urbana-Champaign, Department of Electrical and Computer Engineering**

*Senior Thesis Researcher, ECE 496/499*

Oct. 2022 - Apr. 2023, Urbana, IL, USA

- Developed an approach for localizing errors in LC-3 programming assignments by grouping student solutions according to their structural design patterns.
- Represented program structure using simplified control-flow graphs and investigated automated graph comparison to identify the design approach behind an incorrect solution.
- Explored approach-aware feedback generation to support instruction and debugging in the ECE 220 course.

## **Zhejiang University, NESA Laboratory**

*Undergraduate Research Assistant*

Jul. 2022 - Sep. 2022

- Studied the foundations and implementation of Boolean satisfiability (SAT) and satisfiability modulo theories (SMT) solvers.
- Conducted a source-code-level analysis of MiniSAT and a preliminary analysis of STP to understand their core solving workflows and implementation structures.

## **University of Illinois Urbana-Champaign, SPRAI**

*Undergraduate Research Assistant, Funky Cloud Project*

May 2022 - Aug. 2022, Urbana, IL, USA

- Used Ghidra for static binary analysis on 15 different libraries, including the identification and application of function signatures.
- Curated operating systems and executables across multiple architectures and built test benches for common libraries on the target platforms.

# **Teaching Experience**

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## **Tsinghua University, Institute for Network Sciences and Cyberspace**

*Teaching Assistant, Internet Engineering Design*

Fall 2024, Beijing, China

## **Zhejiang University, ZJUI Institute**

*Teaching Assistant, ECE 220: Computer Systems & Programming*

Fall 2022, Haining, Zhejiang, China

**Zhejiang University, ZJUI Institute**

*Teaching Assistant, MATH 241: Calculus III*

Fall 2021, Haining, Zhejiang, China

## **HONORS & AWARDS**

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- **Graduated with Highest Honors**, Department of Electrical and Computer Engineering, University of Illinois Urbana-Champaign (May 2023)
- **First-Class Scholarship**, Zhejiang University (Jan. 2022)
- **First Prize**, National Mathematics Competition (*Preliminary Round*) (Dec. 2021)
- **Honorable Mention**, Mathematical Contest in Modeling (Apr. 2021)
- **Second-Class Scholarship**, Zhejiang University (Dec. 2020)
- **Second Prize**, National Mathematics Competition (*Preliminary Round*) (Dec. 2020)
- **Second Prize**, International Dragon Boat Competition (Oct. 2020)